Optimisation project

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1 Carbon Capture and Transport Network Model

The model determines the operation of carbon capture sites, the deployment of pipelines, and the transport of captured carbon to storage sites by truck, rail or pipeline.

1.1 Notation

Sets

- $\mathcal{T} = \{0, \ldots, H\}$: set of planning periods, where $H > 0$ is the final period.
- N: set of all nodes.
- N_c: set of capture sites.
- N_t: set of transfer sites.
- N_s: set of storage sites.
- $\mathcal{L} = \{\text{truck}, \text{rail}, \text{pipe}\}$: set of transport modes, indexed by l.

The node sets form a partition of N: $N = N_c \cup N_t \cup N_s$.

Decision Variables

- $F^l_{ij,t} \geq 0$: quantity of carbon transported from node i to node j by transport mode l during period t.
- $S_{ij,t} \geq 0$: pipeline capacity between nodes i and j in period t.
- $w_{ij,t} = \begin{cases} 1 & \text{if a pipeline exists between } i \text{ and } j \text{ in period } t, \\ 0 & \text{otherwise.} \end{cases}$
- $c_{i,t} = \begin{cases} 1 & \text{if carbon capture operates at site } i \text{ in period } t, \\ 0 & \text{otherwise.} \end{cases}$

The implementation defines $c_{i,t}$ for every node, but only variables corresponding to capture sites are used.

Parameters

- L_{ij}: distance between nodes i and j.
- $r_{ij} \in \{0, 1\}$: indicator of an existing rail connection from node i to node j.
- β^l_0: transport cost per unit of carbon and per unit of distance for transport mode l.
- β^l_1: additional transport cost per unit of carbon for transport mode l.
- α_0: pipeline capacity cost coefficient.
- α_1: pipeline length cost coefficient.

- γ^l: emissions per unit of carbon transported and per unit of distance for transport mode l.
- E_i: emissions generated by capture site i per period before capture.
- η: fraction of emissions captured at an operating capture site.
- $b_{i,t}$: storage availability indicator.
- M_1: upper bound on pipeline capacity and on rail flow along an arc in one period.
- M_2: upper bound on the inflow to a storage site in one period.
- Q_{init}: initial bound on transport emissions minus captured emissions.
- Q_{target}: final bound on transport emissions minus captured emissions.

1.2 Objective Function

$$\min \quad \sum_{t\in\mathcal{T}}\sum_{i\in N}\sum_{j\in N}\sum_{l\in\mathcal{L}}\left(\beta_0^l L_{ij}+\beta_1^l\right)F_{ij,t}^l+\sum_{t\in\mathcal{T}}\sum_{i\in N}\sum_{j\in N}\left(\alpha_0 S_{ij,t}+\alpha_1 L_{ij} w_{ij,t}\right). \tag{1}$$

Meaning: Minimise the total transport cost and the pipeline capacity and length costs over the planning horizon.

Note: As implemented, pipeline costs are charged in every period and include both (i,j) and (j,i). For symmetric distances, each physical pipeline's cost is therefore counted twice per period. These are not one-time construction costs.

1.3 Constraints

Pipeline capacity symmetry

$$S_{ij,t}=S_{ji,t} \qquad \forall i,j\in N,\ t\in\mathcal{T}. \tag{2}$$

Meaning: The two directions of a pipeline have the same installed capacity.

Non-decreasing pipeline capacity

$$S_{ij,t+1}\geq S_{ij,t} \qquad \forall i,j\in N,\ t=0,\dots,H-1. \tag{3}$$

Meaning: Installed pipeline capacity cannot decrease over time.

Persistence of installed pipelines

$$w_{ij,t+1}\geq w_{ij,t} \qquad \forall i,j\in N,\ t=0,\dots,H-1. \tag{4}$$

Meaning: Once a pipeline is installed, it remains available in subsequent periods.

Capacity increases only at installation

$$\sum_{t=0}^{H-1}\left(S_{ij,t+1}-S_{ij,t}\right)=\sum_{t=0}^{H-1}\left(S_{ij,t+1}-S_{ij,t}\right)\left(w_{ij,t+1}-w_{ij,t}\right) \qquad \forall i,j\in N. \tag{5}$$

Meaning: Since capacity is non-decreasing and pipeline existence is binary and non-decreasing, capacity may increase only when a pipeline changes from absent to installed. A pipeline already present in period 0 cannot subsequently expand. This constraint is quadratic.

Shared bidirectional pipeline capacity

$$F_{ij,t}^{\text{pipe}}+F_{ji,t}^{\text{pipe}}\leq S_{ij,t} \qquad \forall i,j\in N:\ i<j,\ t\in\mathcal{T}. \tag{6}$$

Meaning: The combined flow in both directions cannot exceed the capacity of the pipeline.

Bidirectional pipeline installation

$$w_{ij,t} = w_{ji,t} \qquad \forall i, j \in N : \ i < j, \ t \in \mathcal{T}. \tag{7}$$

Meaning: A pipeline is installed simultaneously for both directions.

No pipeline installation self-loops

$$w_{ii,t} = 0 \qquad \forall i \in N, \ t \in \mathcal{T}. \tag{8}$$

Meaning: A pipeline cannot connect a node to itself.

Pipeline capacity requires installation

$$S_{ij,t} \leq M_1 w_{ij,t} \qquad \forall i, j \in N, \ t \in \mathcal{T}. \tag{9}$$

Meaning: Pipeline capacity must be zero when the pipeline is absent, and cannot exceed M_1 when it exists. Together with the no-self-loop condition, this also implies $S_{ii,t} = 0$.

Rail availability

$$F^{\text{rail}}_{ij,t} \leq M_1 r_{ij} \qquad \forall i, j \in N, \ t \in \mathcal{T}. \tag{10}$$

Meaning: Rail transport is permitted only along existing rail connections.

No transport self-loops

$$F^{l}_{ii,t} = 0 \qquad \forall i \in N, \ t \in \mathcal{T}, \ l \in \mathcal{L}. \tag{11}$$

Meaning: Carbon cannot be transported from a node back to the same node by any transport mode.

Flow balance at capture sites

$$\sum_{j \in N} \sum_{l \in \mathcal{L}} F^{l}_{ji,t} - \sum_{j \in N} \sum_{l \in \mathcal{L}} F^{l}_{ij,t} = -\eta E_i c_{i,t} \qquad \forall i \in N_c, \ t \in \mathcal{T}. \tag{12}$$

Meaning: The total quantity leaving a capture site equals the total quantity entering it plus the carbon captured at that site. When capture is inactive, the site may still serve as a transit node.

Flow balance at transfer sites

$$\sum_{j \in N} \sum_{l \in \mathcal{L}} F^{l}_{ji,t} - \sum_{j \in N} \sum_{l \in \mathcal{L}} F^{l}_{ij,t} = 0 \qquad \forall i \in N_t, \ t \in \mathcal{T}. \tag{13}$$

Meaning: The total quantity entering a transfer site equals the total quantity leaving it. Carbon may change transport mode at the site.

Storage availability

$$\sum_{j \in N} \sum_{l \in \mathcal{L}} F^{l}_{ji,t} \leq M_2 a_{i,t} \qquad \forall i \in N_s, \ t \in \mathcal{T}. \tag{14}$$

Meaning: A storage site receives no carbon before it becomes available. Once available, it can receive at most M_2 units of carbon per period.

No outgoing flow from storage sites

$$\sum_{j \in N} \sum_{l \in \mathcal{L}} F^{l}_{ij,t} = 0 \qquad \forall i \in N_s, \ t \in \mathcal{T}. \tag{15}$$

Meaning: Storage sites are final destinations. Carbon entering storage cannot leave it.

Emissions reduction trajectory

$$\begin{aligned}\sum_{i\in N}\sum_{j\in N}\sum_{l\in\mathcal{L}}\gamma^{l}L_{ij}F^{l}_{ij,t}-\sum_{i\in N_c}\eta E_i c_{i,t}\\ \leq Q_{\text{init}}+(Q_{\text{target}}-Q_{\text{init}})\,\frac{t}{H}\qquad\forall t\in\mathcal{T}.\end{aligned}\tag{16}$$

Meaning: Transport emissions minus captured emissions must remain below a linearly interpolated bound. If $Q_{\text{target}} < Q_{\text{init}}$, the bound becomes more restrictive over time.

Note: The left-hand side represents an emissions change relative to the uncaptured baseline, not absolute total emissions. Negative values represent net reductions.

Variable domains

$$F^{l}_{ij,t}\geq 0 \qquad \forall i,j\in N,\ t\in\mathcal{T},\ l\in\mathcal{L},\tag{17}$$

$$S_{ij,t}\geq 0 \qquad \forall i,j\in N,\ t\in\mathcal{T},\tag{18}$$

$$w_{ij,t}\in\{0,1\} \qquad \forall i,j\in N,\ t\in\mathcal{T},\tag{19}$$

$$c_{i,t}\in\{0,1\} \qquad \forall i\in N,\ t\in\mathcal{T}.\tag{20}$$

Meaning: Flows and capacities are continuous and nonnegative. Pipeline existence and capture operation are binary decisions.

Model scope The formulation is a mixed-integer quadratically constrained model. Capture operation may switch on or off between periods; only pipeline installation is persistent. Storage has a per-period inflow bound, but no cumulative storage-capacity constraint is imposed.